

Proof of Commitment: Consensus Through Sustained Participation

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Abstract—Permissionless consensus has traditionally derived influence from accumulated resources, most notably computational power in Proof of Work and financial stake in Proof of Stake. While effective for Sybil resistance, these approaches tie consensus influence to resource accumulation, creating persistent incentives for concentration.

This paper explores an alternative foundation in which influence emerges from sustained participation over time. We present *Proof of Commitment* (PoCmt), a permissionless consensus mechanism where validators gradually build a commitment score through recurring protocol engagement across consecutive time windows. Block-production opportunities are allocated according to this accumulated commitment, allowing influence to develop progressively through continued participation rather than through the acquisition of external resources.

PoCmt introduces a participation-based model in which influence is earned independently by each identity over time, making long-term participation the primary source of consensus power. Building on this model, we construct a commitment-weighted leader election mechanism and prove backbone-style safety, liveness, and commitment-proportional fairness under partial synchrony. Security derives from structural participation constraints embedded in the participation model itself.

Seven simulation experiments demonstrate behavior consistent with the theoretical analysis across churn, offline, and adversarial environments. The results show that influence formation is governed by sustained participation capacity rather than by identity replication, while commitment dynamics converge to the rates predicted by the model.

These results establish sustained participation as a structurally distinct and formally grounded basis for permissionless consensus—one that decouples influence from resource accumulation and offers a concrete alternative path toward long-term decentralization.

Index Terms—permissionless consensus, sustained participation, commitment score, Sybil resistance, backbone security, decentralization

I. INTRODUCTION

Decentralization is the foundational promise of blockchain. No single entity should dominate; influence should be distributed, and power should not concentrate. Yet the two paradigms that have come to define permissionless consensus—Proof of Work (PoW) [2]–[4] and Proof of Stake (PoS) [5]–[7]—have produced, in practice, the opposite. Mining pools controlling the majority of global hash power [8], and liquid staking protocols accumulating a third of all staked

ETH [11]: these are not accidents of implementation. They are the predictable outcome of building consensus on *parallelizable resources*—hardware and capital that can be freely pooled and redistributed across arbitrarily many identities without proportional recurring cost. Our companion paper [12] examines this structural property of parallelizable resources and shows that protocols built on such a foundation cannot enforce linear Sybil cost, regardless of how their rules are designed. PoCmt addresses this by replacing the resource class itself—from one that can be parallelized to one that cannot.

A. Commitment as a Resource

This paper asks a different question: what if influence required not *owning* something, but *continuously doing* something?

The intuition is simple. A validator that participates in the protocol today—responding to challenges, engaging with the network—and does so again tomorrow, and the day after, and for months thereafter, is doing something that cannot be replicated in bulk. Each act of participation must be performed per identity, per time window. It cannot be stockpiled in advance, outsourced to a single process without limit, or transferred from one identity to another. What makes this resource scarce is not its quantity but its structure: each identity must earn it separately, in each window, and there is no shortcut—no bulk purchase, no delegation, no way to compress weeks of participation into a single transaction.

An entity wishing to sustain influence across s identities under such a model must supply s independent acts of engagement per window—not once, but repeatedly, for as long as influence is desired. The cost scales linearly with identity count and participation horizon, not with the depth of a balance sheet. This is the structural inversion that breaks the concentration dynamic: capital can be pooled; commitment cannot.

We present *Proof of Commitment* (PoCmt), a permissionless consensus mechanism that instantiates this idea formally. Validators accumulate a *Commitment Score* through recurring protocol engagement across consecutive time windows. Block-production rights are allocated proportionally to this score. Influence grows gradually through sustained participation and cannot be acquired instantaneously through capital injection or hardware scaling.

B. This Work

We develop PoCmt from first principles: a formal model of participation as a non-aggregable resource, a complete consensus protocol built on that foundation, and a security analysis

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Simulation code available at: <https://github.com/homayoun1220/pocmt-simulation>.

proving backbone-style safety, liveness, and commitment-proportional fairness under partial synchrony [16]. Security follows from structural participation constraints and an honest channel majority; it does not depend on the scarcity of any aggregable resource class. Seven analytical experiments confirm that the model behaves as the analysis predicts.

C. Contributions

- 1) **Participation-Based Influence Model (Section IV).** A formal model in which consensus influence derives from sustained per-identity participation. We identify the structural properties that make participation non-aggregable and show that the cost of sustaining s identities over T windows grows as $\Omega(sT)$, independently of capital or hardware.
- 2) **Commitment Score Framework (Section V).** A three-component Commitment Score combining participation history (H_v), protocol compliance (P_v), and availability (U_v). We prove that honest aggregate weight exhibits positive drift over adversarial weight under an honest channel majority (Lemma VII.2), with exponential concentration guarantees (Corollary VII.3).
- 3) **PoCmt Protocol (Section VI).** A complete permissionless consensus protocol with VRF-based commitment-weighted leader election, block proposal and validation, multiplicative slashing, and two finality modes.
- 4) **Backbone-Style Security Analysis (Section VII).** Formal proofs of the common-prefix property (Theorem VII.4), liveness (Theorem VII.6), and commitment-proportional fairness (Lemma VII.7) under partial synchrony. Full derivations appear in Appendices A–C.
- 5) **Analytical Validation (Section VIII).** Seven simulation experiments testing phase transition in adversarial leader share, non-amplification under identity replication, sensitivity to the servicing bound τ and honest surplus ε , robustness under churn, offline recovery, and direct measurement of commitment drift.

D. Organization

Section II surveys related work. Section III establishes the system model. Section IV develops the participation resource model. Section V defines the Commitment Score. Section VI presents the protocol. Section VII proves the security results. Section VIII reports simulation experiments. Section IX discusses scope and limitations. Section X concludes.

II. RELATED WORK

A. Accumulated-Resource Consensus

Douceur [1] established that permissionless systems cannot resist Sybil attacks without binding influence to a costly resource. The dominant instantiations tie that resource to accumulated holdings.

Proof of Work. Nakamoto [2] tied block-production probability to computational effort; backbone-style security was established by Garay et al. [3], [4] and extended to partial synchrony by Pass and Shi [17]. Selfish-mining [9] and

stubborn-mining [10] analyses show how rational deviations concentrate influence in practice; Gencer et al. [8] empirically confirmed 53% pool concentration in Bitcoin.

Proof of Stake. Kiayias et al. [5] gave the first provably secure PoS protocol. Ouroboros Praos [6] extended this to adaptive corruption; Genesis [7] addressed dynamic availability; Badertscher et al. [18] gave a UC-composable treatment. Algorand [19] combines stake-weighted sortition with BFT agreement for low-latency finality. Wahrstätter et al. [11] document liquid staking driving a single protocol to over 32% of staked ETH.

Economic and structural limits. Bonneau et al. [20] survey Bitcoin-style incentive and concentration issues. Budish et al. [21] formalize the economic limits of capital-based security. Lewis-Pye and Roughgarden [22] characterize which consensus properties are achievable under different resource and participation models. Our companion paper [12] establishes the structural result underlying PoCmt: any consensus protocol whose weighting resource can be freely aggregated admits sublinear Sybil cost, regardless of protocol rules. PoCmt responds by replacing the resource class rather than adjusting the protocol.

B. Sequential Work and Time-Based Resources

Proofs of Sequential Work [14], [29] and Verifiable Delay Functions [13], [30] enforce wall-clock lower bounds on individual computations and support delay-based randomness beacons [31]. These are useful building blocks but do not resolve the aggregation problem: an adversary sustaining s identities simply provisions s independent machines and evaluates s delay functions in parallel. The dominant cost remains capital expenditure on hardware, not elapsed time. PoCmt's participation model addresses this gap directly by capping the number of identities a single process can service per window, independently of hardware investment.

C. Alternative Resource Models

Proof-of-Space [32] derives influence from disk capacity; Proof-of-Elapsed-Time [33], [34] relies on trusted execution environments; Proof-of-Useful-Work [35] ties influence to externally valuable computation. These approaches reduce energy consumption or hardware barriers relative to PoW, but their underlying resources remain divisible and transferable, and therefore remain subject to the aggregation dynamics documented in [12].

Graph-based Sybil defenses—SybilGuard [36] and SybilInfer [37]—constrain Sybil identity placement via social network structure. They address where adversarial identities appear, not the recurring cost of sustaining them, and are complementary to PoCmt rather than competitive.

D. Participation and Identity-Binding Mechanisms

Proof-of-Personhood proposals [41], [42] and deployed systems—Worldcoin [43], BrightID [44], Gitcoin Passport [45], PoPCoin [46]—increase the cost of *creating* an identity through biometric or social ceremony. Human-work proposals [47], [48] replace computational puzzles with cognitive

TABLE I
POSITIONING OF PoCMT RELATIVE TO PRIOR WORK. *Persistent* = RECURRING PER-WINDOW COST. *Non-aggregable* = INFLUENCE CANNOT BE POOLED ACROSS IDENTITIES. *Solver-agnostic* = NO HUMAN-SUPERIORITY ASSUMPTION.

Mechanism	Persistent	Non-aggregable	Solver-agnostic
PoW [2]	✓	×	✓
PoS [5]	✓	×	✓
VDF [13]	✓	×	✓
PoP [41]	×	×	×
PoPCoin [46]	✓	×	×
Human-Work [47]	×	Partial	×
CAPTCHA [38]	×	×	×
PoCmt (this work)	✓	✓	✓

tasks. CAPTCHA [38] gates account creation by exploiting perception gaps between humans and solvers.

PoCmt differs from all of these in two respects. First, it enforces a *recurring* per-window cost rather than a one-time admission cost: an adversary that has established s identities must continue supplying s independent participation acts per window to retain influence. Second, PoCmt makes no assumption about human superiority over automated solvers; security derives from structural properties of the participation resource, not from any human-versus-machine capability gap. Table I positions PoCmt relative to the prior landscape.

E. Formal Consensus Security

PoCmt adopts the partial synchrony model of Dwork, Lynch, and Stockmeyer [16] and the backbone proof methodology of Garay et al. [3], [4]. Pass and Shi [27] introduced the sleepy model capturing dynamic validator availability, directly relevant to PoCmt’s churn analysis. Ren [28] provided tighter confirmation-depth bounds for Nakamoto-style protocols. PBFT [23] and HotStuff [24], [25] inform the optional BFT finality gadget. The proof novelty in PoCmt lies not in the backbone methodology but in the arguments connecting participation dynamics (Lemma VII.2) to classical chain properties (Theorem VII.4).

III. SYSTEM MODEL

Table II summarises the core notation.

A. Execution Model

Protocol execution is *round-based*. Time proceeds in discrete *rounds* $r = 1, 2, \dots$ of fixed wall-clock duration $\delta > 0$. Each active validator receives all messages delivered in round r , performs local computation, and sends at most one message to each other validator.

Definition III.1 (Epoch and Participation Window). A consensus epoch t consists of $R_e \in \mathbb{N}_{>0}$ consecutive rounds. A participation window d consists of $E \in \mathbb{N}_{>0}$ consecutive epochs ($R_w = E \cdot R_e$ rounds). We write $\text{ep}(r)$ and $\text{win}(r)$ for the epoch and window containing round r , and $d(t) = \lceil t/E \rceil$.

The two-timescale structure is necessary: consensus messages are exchanged at epoch granularity while participation challenges are scored at window granularity, keeping participation-based influence constant within a window and preventing rapid amplification through short bursts. Window boundaries are globally computable from the protocol transcript without trusted clock infrastructure.

B. Network Model

We adopt the *partial synchrony* model of Dwork, Lynch, and Stockmeyer [16].

Definition III.2 (Partial Synchrony — DLS Model). *There exists an unknown Global Stabilization Time $\text{GST} \geq 0$ such that:*

(N1) *For $r < \text{GST}$, the adversary may delay, reorder, or suppress messages, subject to eventual delivery by round $\text{GST} + \Delta_e$.*

(N2) *For $r \geq \text{GST}$, every honest message sent in round r is received by all honest validators by round $r + \Delta_e$.*

GST is unknown; Δ_e is a public parameter.

Remark III.1. Safety holds for all $r \geq 0$. Liveness holds for $r \geq \text{GST} + \Delta_e$. The adversary may know GST ; our proofs do not rely on the contrary.

C. Participants and Identities

The validator set is *open and dynamic*. Any entity joins by generating a fresh key pair (pk_v, sk_v) under an existentially unforgeable signature scheme [49]; no admission cost or trusted authority is required.

Definition III.3 (Validator Sets). *For round r , let $\mathcal{V}(r) = \mathcal{H}(r) \cup \mathcal{A}(r)$ where $\mathcal{H}(r)$ is the set of honest validators and $\mathcal{A}(r)$ the corrupted set; the two are disjoint.*

We place no a priori bound on $|\mathcal{V}(r)|$ or $|\mathcal{A}(r)|$; security guarantees are stated in terms of *participation capacity* rather than identity count.

D. Adversarial Model

a) *Corruption model.*: The adversary selects $\mathcal{C} \subseteq \mathcal{V}(1)$ before execution begins (static corruption), following the standard backbone assumption [3], [17]. Within each round, the adversary is *rushing*: it observes all honest messages before deciding corrupted validators’ messages. All parties are PPT algorithms parametrised by λ ; security guarantees hold except with probability $\text{negl}(\lambda)$.

Remark III.2 (Adaptive corruption). Adaptive corruption requires the UC framework and significantly increases proof complexity; we defer this to future work (Section IX).

b) *Participation capacity constraints.*: Regardless of computational or financial resources, the adversary cannot:

(C1) credit a valid proof for identity v to any $v' \neq v$;

(C2) carry a proof from window d to any $d' > d$;

(C3) service more than $\tau \in O(1)$ identities per window per independent process.

TABLE II
CORE NOTATION.

Symbol	Meaning
λ	Security parameter
r	Protocol round
t	Consensus epoch (R_e rounds)
d	Participation window (E epochs)
$d(t)$	Window containing epoch t
$\mathcal{V}(r)$	Active validators at round r
$\mathcal{H}(r)$	Honest validators at round r
$\mathcal{A}(r)$	Corrupted validators at round r
v	Validator identity (public key)
τ	Servicing bound: max identities per process per window
$m_{\mathcal{A}}(d)$	Adversarial participation processes in window d
$n_{\mathcal{H}}(d)$	Honest participation channels in window d
$H_v(d)$	Participation commitment of v in window d
$P_v(t)$	Protocol compliance of v at epoch t
$U_v(t)$	Availability of v at epoch t
$CS_v(t)$	Commitment weight of v at epoch t
$W_{\mathcal{H}}(t)$	Total honest commitment weight
$W_{\mathcal{A}}(t)$	Total adversarial commitment weight
$W_{\text{tot}}(t)$	$W_{\mathcal{H}}(t) + W_{\mathcal{A}}(t)$
$C(s, T)$	Min. channel-windows to sustain s identities over T windows
ε	Honest channel surplus (Assumption III.4)
GST	Global Stabilization Time
Δ_e	Post-GST message delay bound (rounds)

These constraints are enforced structurally by the participation resource model (Section IV), independently of adversarial computational power.

E. Honest Participation Assumption

Security is stated over *participation channels* rather than commitment weight to avoid circularity: commitment weight is a protocol output; channels are a structural input constraint.

Assumption III.4 (ε -Honest Channel Majority). *Let $n_{\mathcal{H}}(d)$ and $m_{\mathcal{A}}(d)$ denote, respectively, the number of honest participation channels and adversarial participation processes in window d . There exist $\varepsilon > 0$ and $D_0 \in \mathbb{N}$ such that for all $d \geq D_0$:*

$$n_{\mathcal{H}}(d) \geq (1 + \varepsilon) m_{\mathcal{A}}(d).$$

Remark III.3 (Channel majority versus stake majority). This assumption is weaker than honest-majority assumptions in PoW/PoS in two respects. First, channels cannot be acquired instantaneously: reaching $m_{\mathcal{A}}$ adversarial channels requires operating them across at least D_0 windows, a cost that cannot be front-loaded. Second, the assumption is over processes, not identities: each process services at most τ identities (C3), so unbounded identity creation does not help beyond the $\tau m_{\mathcal{A}}$ ceiling. Lemma VII.2 derives weight majority from these structural constraints rather than asserting it directly.

F. Notation

IV. PARTICIPATION-BASED INFLUENCE MODEL

This section develops the formal model underlying PoCmt's approach to consensus influence. We proceed in four steps:

we define the structural properties that make a participation resource non-aggregable; we model the capacity constraints that follow from those properties; we state the resource assumptions that the deployed mechanism must satisfy; and we derive the cost bound connecting the resource model to the security analysis.

A. Non-Aggregable Participation

In PoW and PoS, a single entity can direct a fixed resource base across arbitrarily many identities without proportional additional cost. The key structural difference in PoCmt is that participation cannot be aggregated in this way. We formalize the conditions that make this so.

Definition IV.1 (Non-Aggregable Participation Resource). *A participation resource $\mathcal{R}$ admits non-aggregable influence if there exists a publicly executable verification predicate $\text{Ver} : \mathcal{ID} \times \mathbb{N} \times \{0, 1\}^* \rightarrow \{0, 1\}$ satisfying the following three properties for all validators v , windows d , and security parameter λ :*

- (R1) Identity binding.** $\text{Ver}(v, d, \pi) = 1$ implies $\text{Ver}(v', d, \pi) = 0$ for all $v' \neq v$, except with probability $\text{negl}(\lambda)$. A valid participation proof cannot be credited to any identity other than the one that produced it.
- (R2) Window locality.** $\text{Ver}(v, d, \pi) = 1 \Rightarrow \text{Ver}(v, d', \pi) = 0$ for all $d' \neq d$. Participation must be renewed each window; proofs cannot be carried forward or reused.
- (R3) Bounded concurrent servicing.** There exists $\tau \in \mathbb{N}_{>0}$ such that any single process p produces at most τ valid proofs across distinct identities per window:

$$|\{v : \text{Ver}(v, d, \pi_p^v) = 1\}| \leq \tau,$$

independently of p 's computational resources or the number of identities it attempts to service.

Resources satisfying (R1)–(R3) are called time-per-identity (TPI) resources.

Properties (R1) and (R2) together prevent influence from being transferred between identities or banked across windows without fresh participation. Property (R3) is the aggregation barrier: a single process can supply participation for at most τ identities per window, so sustaining s identities requires at least $\lceil s/\tau \rceil$ independent processes. Capital and hardware can increase the quality or speed of individual participation but cannot bypass this ceiling.

Remark IV.1 (Contrast with existing resource classes). PoW and PoS fail (R2) and (R3): a fixed resource base is reusable across windows and can be subdivided to service arbitrarily many identities simultaneously. VDFs [13] satisfy a sequential-time constraint within a single execution but fail (R3): an adversary provisions s independent machines and runs s delay functions in parallel at no additional per-identity recurring cost. This class of resource enforces (R3) at the process level, requiring proportionally more independent processes for each additional identity.

Remark IV.2 (Solver agnosticism). Properties (R1)–(R3) constrain the *structure* of participation, not the identity of the

solver. Whether proofs are generated by humans, automated agents, or hybrid systems is orthogonal to the cost analysis; only the process-level throughput ceiling τ matters for security.

B. Participation Capacity Model

We now model the concrete capacity constraints that follow from Definition IV.1 (the participation resource properties).

A *participation process* is an independent computational unit capable of generating valid participation proofs on behalf of one or more identities. The key quantities are:

- τ : the *servicing bound*—the maximum number of distinct identities a single process can validly service per window, enforced by (R3).
- m : the number of independent participation processes an entity operates.
- s : the number of identities the entity wishes to sustain with nontrivial commitment.

The relationship between these quantities determines whether influence can be aggregated. If a single process could service an unbounded number of identities ($\tau \rightarrow \infty$), one entity with one process could sustain arbitrarily many identities—the PoW/PoS regime. Property (R3) caps this: to sustain s identities, an entity needs at least $\lceil s/\tau \rceil$ processes. When $\tau = O(1)$, this means $\Omega(s)$ processes are required, making the cost of sustaining many identities grow proportionally with s .

This capacity model is the bridge between the abstract resource definition and the security analysis: it translates (R3) into a concrete bound on adversarial proof production, which in turn bounds adversarial commitment growth.

C. Participation Resource Assumptions

We assume that the participation resource deployed in PoCmt satisfies three structural properties. This is analogous to assuming a collision-resistant hash function in PoW: the assumption characterizes the required properties of the mechanism without prescribing a specific construction. Deployed participation systems have demonstrated the operational feasibility of recurring per-identity participation bounds at scale [15]; a formal cryptographic construction is an open problem (Section IX-F).

Assumption IV.2 (Participation Resource). *The deployed participation resource satisfies, for all validators v , windows d , and identities $v' \neq v$:*

- (A1) **Identity binding.** *A valid proof for v in window d cannot be credited to any $v' \neq v$, except with probability $\text{negl}(\lambda)$. Enforces (R1).*
- (A2) **Window locality.** *Proofs cannot be precomputed before challenge issuance and expire at the window boundary. Enforces (R2).*
- (A3) **Bounded concurrent servicing.** *There exists $\tau > 0$ such that any single process produces valid proofs for at most τ distinct identities per window, independently of computational power. Enforces (R3).*

Remark IV.3 (Conditionality). Assumption IV.2 plays the same role as the honest-majority assumption in PoS or the collision-resistance assumption in PoW: a structural precondition on the deployment environment, not a consequence of protocol rules. All security results in Section VII are conditional on Assumption IV.2.

Concrete mechanisms satisfying Assumption IV.2 include per-session serialization (one active challenge per session token), device-bound attestation (a hardware enclave enforcing per-device rate limits), and short-deadline per-identity challenges. Each enforces (A3) and represents a different trade-off between τ , participation convenience, and infrastructure dependency.

D. Adversarial Capacity Bound

The following proposition formalizes the capacity model of Section IV-B under Assumption IV.2, and is the key structural result connecting the resource model to the security analysis.

Proposition IV.3 (Adversarial Capacity Bound). *Under Assumption IV.2, if the adversary operates m independent processes in window d , then the total valid participation proofs it produces satisfies:*

$$X_A(d) = \sum_{v \in \mathcal{A}} x_v(d) \leq m \cdot \tau,$$

where $x_v(d) \in \{0, \dots, k(d)\}$ is the count of valid proofs for identity v . Consequently, maintaining full participation across s identities requires $m = \Omega(s)$ concurrent processes, regardless of capital or hardware.

Proof. By (A1), proofs for distinct identities are independent; no proof is credited to more than one identity. By (A2), each proof must be produced online within the window; precomputation is impossible. By (A3), a single process produces at most τ valid proofs per window. Summing over m independent processes: $X_A(d) \leq m \cdot \tau$. For full participation across s identities, $X_A(d) \geq s \cdot k(d)$, giving $m \geq s \cdot k(d)/\tau = \Omega(s)$. $\square$

Corollary IV.4 (Non-Aggregability of Influence). *Under Assumption IV.2, for any number of adversarial identities s : $\Delta W_A(d) = O(m)$, independent of s . Adversarial influence gain per window is bounded by process count, not identity count.*

Proof. By Proposition IV.3, total valid proofs per window $\leq m \cdot \tau$. Since commitment increments are proportional to solved challenges, total H -component growth is $O(m \cdot \tau) = O(m)$, independent of s . $\square$

Corollary IV.4 is the formal statement of non-aggregability: registering additional adversarial identities beyond the process capacity $m \cdot \tau$ yields no additional commitment growth. The adversary's influence is determined entirely by how many independent participation processes it sustains, not by how many identities it registers.

E. Cost Structure

Definition IV.5 (Participation Cost). *The adversarial participation cost $C(s, T)$ is the minimum number of independent participation channel-windows required to sustain nontrivial commitment weight across s distinct identities for T consecutive windows:*

$$C(s, T) = \min_{(m_{\mathcal{A}}(d))_{d=1}^T} \sum_{d=1}^T m_{\mathcal{A}}(d)$$

subject to $|\{v \in \mathcal{A}(d) : H_v(d) \geq H_{\min}\}| \geq s$ for all $d \in [T]$, over all adversarial strategies satisfying constraints (C1)–(C3) of Section III.

Definition IV.6 (Marginal Cost). $\Delta C(s, T) = C(s, T) - C(s-1, T)$ is the marginal cost of sustaining one additional identity over T windows.

Two asymptotic regimes characterize the cost structure. When $C(s, T) = o(sT)$, the marginal cost of an additional identity diminishes as s grows: influence can be aggregated and the protocol is susceptible to concentration. When $C(s, T) = \Omega(sT)$, the marginal cost remains bounded away from zero regardless of scale: influence is non-aggregable. The companion paper [12] establishes that every protocol built on a freely aggregable resource falls into the first regime. Theorem VII.1 (Section VII) establishes that PoCmt, under Assumption IV.2, falls into the second.

V. COMMITMENT SCORE

Unlike PoW—where influence equals current hash rate— or PoS—where it equals current stake—influence in PoCmt accumulates over time and cannot be instantaneously acquired, transferred, or delegated.

A. Commitment State

Each validator v maintains a *commitment state* $S_v(t) = (H_v(t), P_v(t), U_v(t))$ comprising:

- $H_v(t) \geq 0$: *Participation commitment*. Accumulated through per-identity, per-window challenges. The primary non-amortizable component; cannot be pooled or delegated.
- $P_v(t) \geq 0$: *Protocol compliance*. Reflects conformant message exchange and block validation; decreases multiplicatively upon slashing, keeping penalties separate from participation history.
- $U_v(t) \geq 0$: *Availability*. Reflects recent online responsiveness; decays exponentially during offline periods, preventing dormant identities from retaining persistent influence.

B. Update Rules

a) *Participation commitment H_v (window-level)*.: At the start of window d , the participation resource issues $k(d)$ fresh challenges to validator v . Let $x_v(d) \in \{0, \dots, k(d)\}$ be the number of valid responses submitted within deadline Δ_{resp} . At the end of window d :

$$H_v(d+1) = H_v(d) + \kappa_h \cdot x_v(d), \quad \kappa_h > 0. \quad (1)$$

H_v is constant within a window, ensuring participation influence cannot be amplified through short-term bursts.

b) *Protocol compliance P_v (epoch-level)*.: Validators accumulate compliance credit through conformant actions. Upon on-chain verification of a slashable violation:

$$P_v(t) \leftarrow \delta \cdot P_v(t), \quad 0 < \delta < 1. \quad (2)$$

H_v is *never* reduced by slashing: time-per-identity expenditure already incurred cannot be retroactively invalidated.

c) *Availability U_v (epoch-level)*.: If v is unresponsive in epoch t :

$$U_v(t+1) = \lambda \cdot U_v(t), \quad 0 < \lambda < 1. \quad (3)$$

Online, U_v increases by $\mu > 0$, bounded by $U_{\max} = \mu/(1-\lambda)$.

C. Commitment Score

Definition V.1 (Commitment Score).

$$CS_v(t) = \alpha H_v(d(t)) + \beta P_v(t) + \gamma U_v(t), \quad (4)$$

where $\alpha, \beta, \gamma \geq 0$, $\alpha + \beta + \gamma = 1$. Security results hold for any $\alpha, \beta, \gamma > 0$.

Aggregate weights: $W_{\mathcal{H}}(t) = \sum_{v \in \mathcal{H}(t)} CS_v(t)$, $W_{\mathcal{A}}(t) = \sum_{v \in \mathcal{A}(t)} CS_v(t)$, $W_{\text{tot}}(t) = W_{\mathcal{H}}(t) + W_{\mathcal{A}}(t)$.

Remark V.1 (Role of α). Setting α close to 1 maximises Sybil resistance; lower α allows compliance and availability to contribute more, which may suit deployments where participation capacity is naturally constrained.

D. Properties of the Commitment Score

Proposition V.2 (Non-amortizability of CS). *Under Assumption IV.2, for any adversary with m processes and any s : $W_{\mathcal{A}}(d+1) - W_{\mathcal{A}}(d) = O(m)$, independent of s .*

Proof. By Corollary IV.4, total valid proofs $\leq m\tau$ per window; each contributes at most $\alpha\kappa_h$ to CS_v . Since $P_v \leq P_{\max}$ and $U_v \leq U_{\max}$ are finite, $W_{\mathcal{A}}(d+1) - W_{\mathcal{A}}(d) \leq \alpha\kappa_h\tau m + O(1) = O(m)$. $\square$

Proposition V.3 (Honest drift). *Under Assumption III.4 and honest participation, with $n_{\mathcal{H}}$ channels and $m_{\mathcal{A}} \leq n_{\mathcal{H}}/(1+\varepsilon)$:*

$$\mathbb{E}[W_{\mathcal{H}}(d+1) - W_{\mathcal{H}}(d)] \geq \mu_H \cdot n_{\mathcal{H}},$$

where $\mu_H = \alpha\kappa_h k_0 > 0$, $k_0 = \min_d k(d)$.

Proof. Each honest channel completes all $k(d) \geq k_0$ challenges per window, contributing $\geq \alpha\kappa_h k_0 = \mu_H$ per channel. Summing over $n_{\mathcal{H}}$ channels gives the bound. $\square$

Proposition V.4 (Inactivity decay). *If v is offline for k consecutive epochs: $U_v(t+k) \leq U_v(t) \cdot \lambda^k \rightarrow 0$.*

Proof. Direct from (3). $\square$

Propositions V.2–V.4 establish the three guarantees for Section VII: adversarial weight growth is channel-bounded, honest weight grows with channel count, and dormant validators lose influence exponentially.

VI. THE PoCMT PROTOCOL

PoCmt follows the Nakamoto-style architecture [3], [17], replacing hash power or stake with accumulated participation commitment.

A. Leader Election

Each validator v computes at epoch t :

$$r_v(t) = \text{VRF}(\text{sk}_v, t \parallel \text{rand}(t)) \in [0, 1],$$

where $\text{rand}(t)$ is a public randomness beacon [31]. Validator v is eligible to propose if:

$$r_v(t) < \Theta \cdot \frac{CS_v(t)}{\sum_{u \in \mathcal{V}(t)} CS_u(t)},$$

where Θ is calibrated for a constant expected number of eligible leaders per epoch. Because H_v updates only at window boundaries, leader-election probabilities are piecewise constant within each window.

B. Block Proposal and Validation

An elected leader ℓ_t broadcasts block B_t containing: parent hash and epoch t , transaction payload, VRF output with eligibility proof, and leader signature.

Each validator:

- 1) verifies the VRF output and eligibility proof;
- 2) verifies the leader signature;
- 3) validates consistency with the local fork-choice rule;
- 4) validates included transactions;
- 5) detects and reports equivocation.

Blocks failing any check are rejected. Valid slashing proofs are propagated and applied deterministically.

C. Slashing

Slashable behavior: equivocation (two blocks for the same epoch), invalid VRF proof, or misrepresented commitment state. Upon on-chain verification: $P_v \leftarrow \delta P_v$ ($0 < \delta < 1$). H_v is *never* reduced by slashing, preserving the incentive for continued honest engagement.

D. Fork Choice and Finality

a) *Weighted longest-chain mode.*: Validators follow the chain $\mathcal{C}$ maximising: $W(\mathcal{C}) = \sum_{B \in \mathcal{C}} CS_{\text{leader}(B)}(t_B)$. Under weighted honest majority, backbone-style chain-growth, chain-quality, and common-prefix follow from [3], [17].

b) *Optional BFT finality gadget.*: A HotStuff-style gadget [23], [24] finalizes a block once it accumulates $> \frac{2}{3}$ of total commitment weight. Under partial synchrony with $W_A < \frac{1}{3} W_{\text{tot}}$, safety holds unconditionally and liveness holds after GST.

E. Protocol Timeline

Each participation window d :

- 1) The participation resource issues $k(d)$ fresh per-identity challenges.
- 2) Validators submit responses within Δ_{resp} .
- 3) H_v updated via (1).

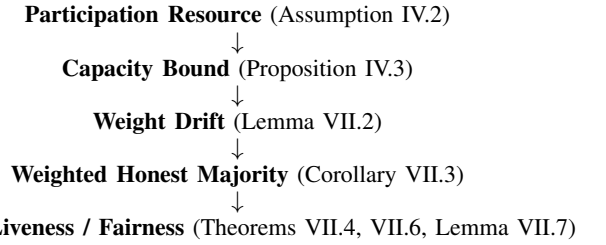
Each epoch t within window d :

- 1) Message exchange; $P_v(t)$ updated.
- 2) Availability updates; $U_v(t)$ computed.
- 3) Scores $CS_v(t)$ recomputed.
- 4) VRF leader election.
- 5) Eligible leaders propose blocks.
- 6) Validators apply fork choice and slashing proofs.

VII. SECURITY ANALYSIS

We establish backbone-style security under partial synchrony [16] in four stages: (i) linear Sybil cost, (ii) commitment drift and stabilization, (iii) safety and liveness, (iv) fairness. Full proofs appear in Appendices A–C.

The logical dependency structure of the analysis is as follows:



Security flows from the structural properties of the participation resource upward to the classical backbone guarantees. Weighted honest majority is not assumed independently; it is derived from the participation resource properties (Assumption IV.2) and the honest channel majority (Assumption III.4).

A. Linear Sybil Cost

Theorem VII.1 (Linear Sybil Cost). *Under Assumption IV.2, sustaining s adversarial identities with nontrivial commitment over T windows requires $C(s, T) = \Omega(sT)$.*

Proof. By Proposition IV.3, each window requires $\Omega(s)$ adversarial channels. With $m = o(s)$ channels, at most $m\tau = o(s)$ identities receive H_v increments; by Corollary IV.4, $\Delta W_A = O(m)$ per window. By Proposition V.3, $\Delta W_H \geq \mu n_H$, so honest weight dominates for $m = o(s)$. Sustaining nontrivial weight over T windows requires $\Omega(s)$ channels per window, giving $C(s, T) = \Omega(sT)$. $\square$

B. Security Assumptions

- **(A1) Partial synchrony.** Definition III.2 holds with unknown GST and known Δ_e .
- **(A3) Honest participation.** After GST, honest validators complete all challenges and remain online except for transient bounded-duration faults.
- **(A4) Resource integrity.** The deployed participation resource satisfies Assumption IV.2 throughout execution.

Remark VII.1 (Weighted honest majority is derived). $W_{\mathcal{A}} < \frac{1}{2}W_{\text{tot}}$ is not assumed independently; it follows from Assumption III.4 via Lemma VII.2 and Corollary VII.3: for all $d \geq D_1$, $W_{\mathcal{H}}(d) > W_{\mathcal{A}}(d)$ with probability $1 - \exp(-\Omega(d))$.

C. Commitment Drift

Define the *commitment gap* $\Delta(d) = W_{\mathcal{H}}(d) - W_{\mathcal{A}}(d)$.

Lemma VII.2 (Quantitative Commitment Drift). *Let $k_0 = \min_d k(d)$, $\mu = \alpha\kappa_h k_0$, $c = \tau/k_0$. Under (A3) and Assumption III.4 with $n_{\mathcal{H}} \geq (1 + \varepsilon)m_{\mathcal{A}}$, for all $d \geq D_0$:*

$$\mathbb{E}[\Delta(d+1) \mid \mathcal{F}_d] \geq \Delta(d) + \gamma,$$

where $\gamma = \mu\varepsilon m_{\mathcal{A}} > 0$.

Proof sketch. Honest channels contribute $\geq \mu$ per window; the adversary generates at most $\mu c m_{\mathcal{A}}$ total growth. Net drift $\geq \mu(1 + \varepsilon - c)m_{\mathcal{A}} =: \gamma > 0$. See Appendix A for the full proof. $\square$

Corollary VII.3 (Exponential Stabilization). *Let B bound $|\Delta(d+1) - \Delta(d)|$ a.s. For all $d \geq D_0$, $\xi > 0$:*

$$\Pr[\Delta(d) \leq \Delta(D_0) + (d - D_0)\gamma - \xi] \leq \exp\left(-\frac{2\xi^2}{(d - D_0)B^2}\right).$$

Once $\Delta(D_0) > 0$, adversarial takeover probability decays exponentially in d .

Proof sketch. $X_i = \Delta(D_0 + i) - \Delta(D_0) - i\gamma$ is a submartingale by Lemma VII.2; Azuma–Hoeffding applied to $Y_i = -X_i$ gives the bound. See Appendix B. $\square$

D. Safety

By Corollary VII.3, for all $d \geq D_1$: $p_{\mathcal{H}}(t) \geq p_{\min} = (1 + \eta)/(2 + \eta) > \frac{1}{2}$ with probability $1 - \exp(-\Omega(d))$.

Theorem VII.4 (Common-Prefix Property). *Under (A1) and Assumption III.4, for all epochs in windows $d \geq D_1$ and any depth k :*

$$\Pr[\text{common-prefix violation at depth } k] \leq e^{-2\delta k} = \text{negl}(k),$$

where $\delta = p_{\min} - \frac{1}{2} > 0$.

Proof sketch. With $p = p_{\min} > \frac{1}{2}$, by the ballot-problem argument $\Pr[F_k] \leq (q/p)^k \leq e^{-2\delta k}$. See Appendix C for the full argument including pre-stabilization and slashing. $\square$

Theorem VII.5 (Safety Under BFT Finality). *If $W_{\mathcal{A}} < \frac{1}{3}W_{\text{tot}}$, two conflicting blocks cannot both be finalized under a $\frac{2}{3}$ -weight quorum rule.*

Proof. Two quorums each exceeding $\frac{2}{3}$ intersect in $> \frac{1}{3}$ of weight. Since $W_{\mathcal{A}} < \frac{1}{3}W_{\text{tot}}$, at least one honest validator lies in the intersection and cannot sign both [23], [24]. $\square$

TABLE III
SECURITY GUARANTEES OF PoCMT.

Property	Condition	Result
Linear Sybil cost	Asm. IV.2	Thm. VII.1
Drift stabilization	$n_{\mathcal{H}} \geq (1 + \varepsilon)m_{\mathcal{A}}$	Cor. VII.3
Common prefix	$W_{\mathcal{A}} < \frac{1}{2}W_{\text{tot}}$	Thm. VII.4
BFT safety	$W_{\mathcal{A}} < \frac{1}{3}W_{\text{tot}}$	Thm. VII.5
Liveness	Partial sync, ≥ 1 honest online	Thm. VII.6
Fairness	$CS_u/CS_v \rightarrow r$	Lem. VII.7

E. Liveness

Theorem VII.6 (Liveness). *Under (A1), with at least one honest validator online after GST, the chain grows without bound and every valid transaction is eventually confirmed.*

Proof. Step 1. By Corollary VII.3, $\exists D_1, \eta > 0$: $W_{\mathcal{H}} \geq (1 + \eta)W_{\mathcal{A}}$ for $d \geq D_1$, giving $p_{\min} = (1 + \eta)/(2 + \eta) > \frac{1}{2}$.

Step 2. Honest blocks in n epochs dominate $\text{Binomial}(n, p_{\min})$. Chernoff: $\Pr[\text{honest blocks} < (p_{\min} - \zeta)n] \leq e^{-2\zeta^2 n}$ for any $\zeta > 0$.

Step 3. A transaction is confirmed within geometrically distributed time: $(1 - p_{\min})^k \leq e^{-p_{\min}k} \rightarrow 0$. $\square$

F. Fairness

Lemma VII.7 (Commitment-Proportional Fairness). *Let u, v be honest validators with $CS_u(t)/CS_v(t) \rightarrow r$. Then $L_u(n)/L_v(n) \xrightarrow{a.s.} r$, where $L_v(n)$ counts v 's leader elections in n epochs.*

Proof. VRF independence makes each $L_v(n)$ a sum of Bernoullis. By the Strong Law of Large Numbers, $L_v(n)/n \xrightarrow{a.s.} \bar{p}_v$. $CS_u/CS_v \rightarrow r$ implies $p_u(t)/p_v(t) \rightarrow r$, so $\bar{p}_u/\bar{p}_v = r$ and $L_u(n)/L_v(n) \xrightarrow{a.s.} r$. $\square$

G. Summary

VIII. ANALYTICAL VALIDATION

This section reports eight experiments that validate the theoretical predictions of Sections IV–VII. Experiments 1–7 isolate individual model parameters and confirm that simulation output matches the analytical predictions; Experiment 8 compares influence concentration across PoCmt, PoW, and PoS under a unified population model, directly measuring the paper's central claim.

All experiments are implemented in a modular Python simulation suite. The PoCmt engine implements the window-level update rules of Section V exactly; honest validators complete participation challenges with rates drawn from $\text{Beta}(5, 2)$ (mean ≈ 0.71), modeling realistic heterogeneity in validator reliability. PoW influence is distributed as $\text{Pareto}(1.16)$, consistent with empirical Bitcoin mining pool data [8]; PoS influence follows $\text{LogNormal}(0, 1.5)$, consistent with observed Ethereum stake distributions [11]. All experiments run for $W = 300$ participation windows with $N = 100$ independent random seeds. Default parameters: $\alpha = 0.7$, $\kappa_h = 1$, $k(d) = 1$, $\tau = 1$, $\lambda = 0.9$, $\mu_U = 0.1$. Each experiment states its hypothesis before reporting results.

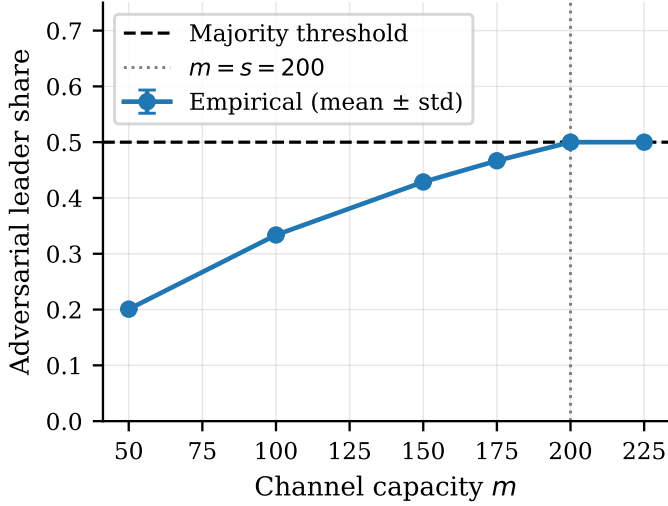


Fig. 1. Adversarial leader share vs. channel capacity m ($s = 200$, $N = 100$ runs, $W = 300$ windows; error bars show ± 1 std). The transition is sharp at $m = s = 200$. Dashed line: majority threshold (0.5). Dotted vertical: $m = s$.

TABLE IV
STEADY-STATE ADVERSARIAL LEADER SHARE AT WINDOW 300
($s = n_{\mathcal{H}} = 200$, MEAN OVER 100 RUNS).

m	Leader share
50	0.20
100	0.33
150	0.43
175	0.47
200	0.50
225	0.50

A. Experiment 1: Phase Transition in Leader Share

Setup. Fix $s = n_{\mathcal{H}} = 200$; vary $m \in \{50, 100, 150, 175, 200, 225\}$.

Hypothesis. By Theorem VII.1, adversarial leader share is governed by m , not s . A sharp phase transition should occur at $m = s = 200$, with share remaining well below majority for $m < s$.

Result. Figure 1 shows a sharp transition at $m = s = 200$: share rises from 0.20 at $m = 50$ to 0.50 at $m = 200$, then saturates. Table IV reports steady-state values.

B. Experiment 2: Non-Amplification Under Identity Replication

Setup. Fix $m = 100$, $n_{\mathcal{H}} = 200$; vary $s \in \{50, 200, 1,000, 5,000\}$.

Hypothesis. By Corollary IV.4, adversarial influence gain per window is $O(m)$, independent of s . Leader share should remain flat as s grows.

Result. Figure 2 shows leader share remaining flat from $s = 50$ to $s = 5,000$ under fixed $m = 100$ (values: 0.20, 0.33, 0.34, 0.35). Identity replication without additional participation processes yields negligible influence gain.

C. Experiment 3: Sensitivity to Servicing Bound τ

Setup. Fix $m = 100$, $s = n_{\mathcal{H}} = 200$; vary $\tau \in \{1, 2, 5, 10\}$.

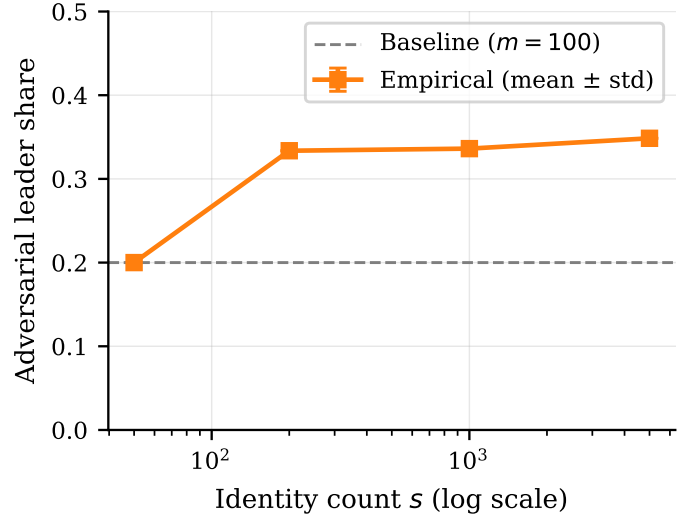


Fig. 2. Adversarial leader share vs. identity count s under fixed $m = 100$ ($N = 100$ runs, $W = 300$ windows, log-scale x -axis; error bars show ± 1 std). Share is essentially invariant as s grows two orders of magnitude, validating Corollary IV.4.

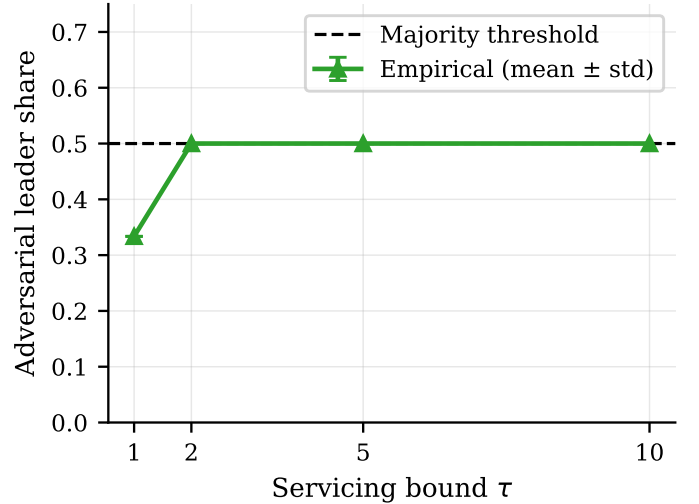


Fig. 3. Adversarial leader share vs. servicing bound τ ($m = 100$, $s = n_{\mathcal{H}} = 200$, $N = 100$ runs; error bars show ± 1 std). Share reaches majority at $\tau = 2$ ($m\tau = 200 = s$), motivating tight challenge design.

Hypothesis. Effective adversarial capacity is $m\tau$ (Proposition IV.3). Leader share should grow with τ , reaching majority when $m\tau \geq s$.

Result. Figure 3 confirms share rises from 0.33 at $\tau = 1$ to 0.50 at $\tau = 2$ (where $m\tau = s$) and saturates thereafter. This establishes $\tau = 1$ as the critical design target for the participation resource.

D. Experiment 4: Sensitivity to Honest Channel Surplus ε

Setup. Fix $m_{\mathcal{A}} = 100$; vary $n_{\mathcal{H}} \in \{101, 110, 120, 150, 200\}$, corresponding to $\varepsilon \in \{0.01, 0.1, 0.2, 0.5, 1.0\}$.

Hypothesis. By Lemma VII.2, drift rate γ grows linearly with ε . Larger ε should yield lower adversarial share at window 300.

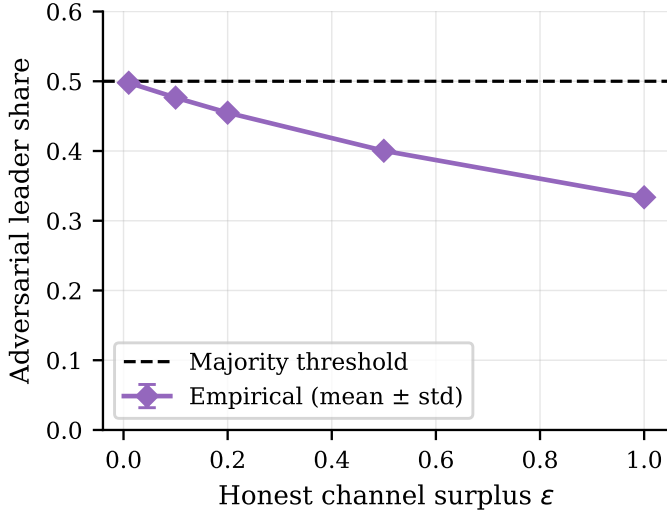


Fig. 4. Adversarial leader share vs. honest channel surplus ε at window 300 ($m_A = 100$, $s = 200$, $N = 100$ runs; error bars show ± 1 std). Share decreases monotonically with ε , validating Lemma VII.2.

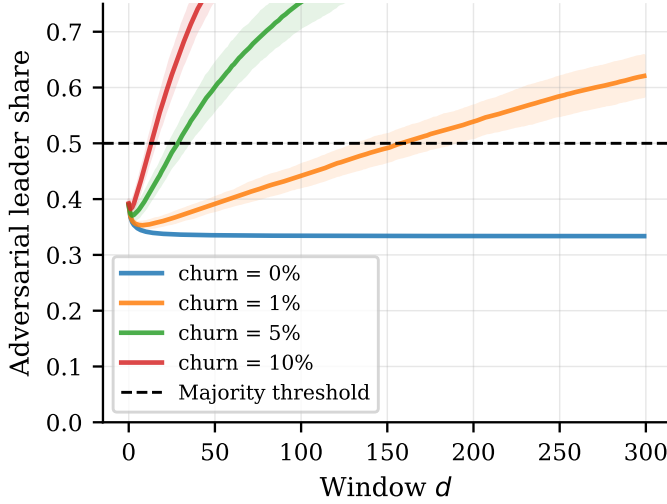


Fig. 5. Adversarial leader share over 300 windows under symmetric validator churn ($m_A = 100$, $n_H = 200$, $N = 100$ runs; shaded bands show ± 1 std). The protocol remains stable at churn $\leq 5\%$.

Result. Figure 4 shows share decreasing monotonically from 0.50 at $\varepsilon = 0.01$ to 0.33 at $\varepsilon = 1.0$, consistent with the predicted drift rate $\gamma = \mu\varepsilon m_A$.

E. Experiment 5: Validator Churn

Setup. Symmetric join/leave rates $p \in \{0\%, 1\%, 5\%, 10\%\}$ per window; $n_H = 200$, $m_A = 100$, $s = 200$.

Hypothesis. By Corollary VII.3, exponential drift concentration should maintain honest majority under moderate churn; heavy churn introduces transient fluctuations but majority is preserved.

Result. Figure 5 confirms stability under churn $\leq 5\%$; at 10%, honest majority is maintained despite transient oscillations, consistent with Corollary VII.3.

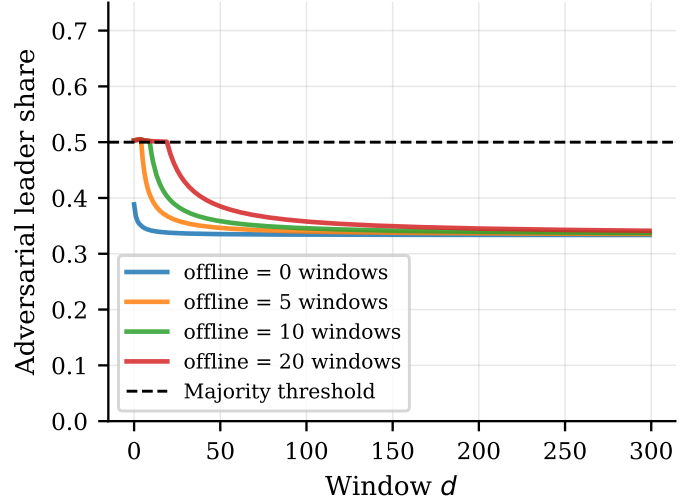


Fig. 6. Adversarial leader share with k initial offline windows ($n_H = 200$, $m_A = 100$, $N = 100$ runs; shaded bands show ± 1 std). Honest share recovers rapidly after the offline period.

F. Experiment 6: Offline Nodes and Recovery

Setup. First $k \in \{0, 5, 10, 20\}$ windows with 50% of honest validators offline; $n_H = 200$, $m_A = 100$, $s = 200$.

Hypothesis. The availability component U_v decays exponentially during offline periods (Proposition V.4). After validators return online, honest share should recover rapidly as U_v rebuilds and H_v resumes accumulating.

Result. Figure 6 shows rapid recovery after offline periods of up to 20 windows. The U_v mechanism prevents dormant adversarial identities from retaining stale influence, limiting the impact of transient honest unavailability.

G. Experiment 7: Commitment Drift

Setup. Measure $\Delta(d) = W_H(d) - W_A(d)$ for $n_H \in \{120, 150, 200, 300\}$, $m_A = 100$, $s = 200$.

Hypothesis. By Lemma VII.2, $\Delta(d)$ should grow linearly with slope $\gamma = \mu\varepsilon m_A$, where $\varepsilon = n_H/m_A - 1$.

Result. Figure 7 shows $\Delta(d)$ growing linearly in all four configurations, with empirical slopes closely matching the theoretical predictions (dashed lines). The match improves with larger ε , consistent with the asymptotic nature of the bound.

H. Experiment 8: Influence Concentration Comparison

Setup. We compare influence concentration across three mechanisms under a unified population of $N = 200$ validators. PoW influence is Pareto(1.16)-distributed; PoS influence is LogNormal(0, 1.5)-distributed. Distribution parameters were selected to approximately match the empirically observed concentration levels reported in [8] and [11], respectively. PoCmt influence is derived from the simulation engine with $m_A = 100$ adversarial processes and heterogeneous honest participation rates drawn from Beta(5, 2) (mean ≈ 0.71), modeling realistic validator reliability. All three are evaluated over 100 independent runs. Concentration is measured by

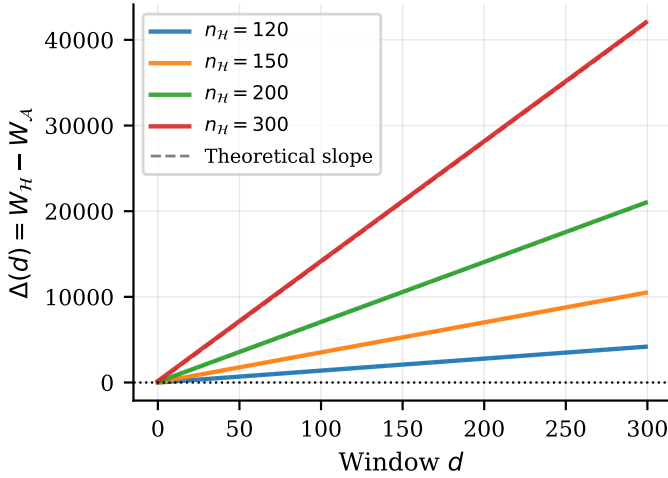


Fig. 7. Commitment gap $\Delta(d) = W_H - W_A$ over 300 windows for four honest channel counts ($m_A = 100$, $N = 100$ runs; shaded bands show ± 1 std). Solid: empirical mean. Dashed: theoretical slope $\gamma = \mu\epsilon m_A$ (Lemma VII.2).

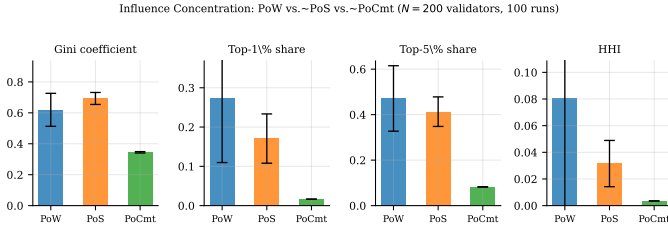


Fig. 8. Influence concentration metrics for PoW, PoS, and PoCmt ($N = 200$ validators, 100 runs; error bars show ± 1 std). PoCmt achieves substantially lower concentration on all four metrics.

Gini coefficient, top-1% and top-5% influence shares, and the Herfindahl–Hirschman Index (HHI).

Hypothesis. PoCmt’s non-aggregability property (Corollary IV.4) should yield substantially lower concentration than PoW and PoS, in which influence is freely aggregable.

Result. Figures 8 and 9 and Table V report the results. PoCmt achieves a Gini coefficient of 0.345 ± 0.005 , compared to 0.619 ± 0.107 for PoW and 0.693 ± 0.039 for PoS. The top-1% influence share is 1.6% under PoCmt, versus 27.3% under PoW and 17.1% under PoS. The HHI is 0.004 for PoCmt, compared to 0.080 for PoW and 0.032 for PoS. The Lorenz curve for PoCmt lies substantially closer to the equality diagonal than those for PoW and PoS, confirming that the participation-based influence model produces structurally lower concentration under equivalent validator populations.

I. Summary

Across all eight experiments the simulation confirms the theoretical predictions. Experiments 1–7 establish that adversarial influence scales with participation capacity rather than identity count, that commitment drift stabilizes at the predicted rate, and that the protocol remains stable under moderate churn and transient offline periods. Experiment 8 establishes the central comparative claim: under equivalent validator populations, PoCmt produces substantially lower influence concentration

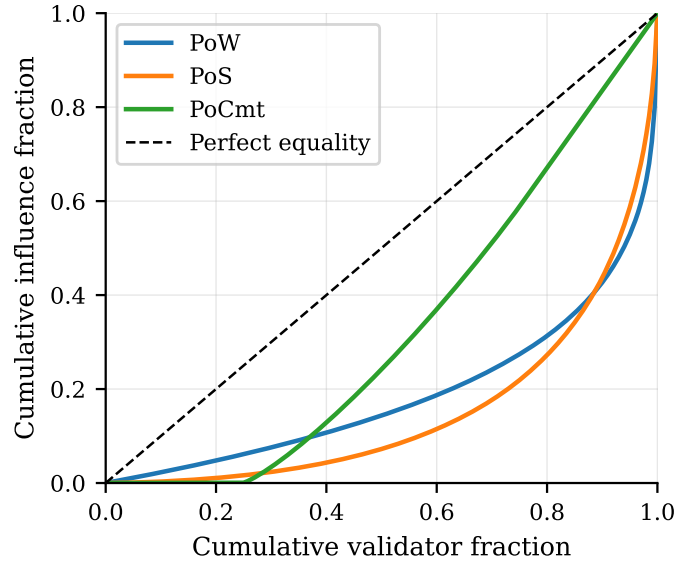


Fig. 9. Lorenz curves for PoW, PoS, and PoCmt ($N = 200$ validators, mean over 100 runs). The PoCmt curve lies substantially closer to the perfect-equality diagonal, confirming lower structural concentration.

TABLE V
INFLUENCE CONCENTRATION METRICS FOR PoW, PoS, AND PoCMT ($N = 200$ VALIDATORS, MEAN $\pm$ STD OVER 100 RUNS). LOWER IS LESS CONCENTRATED.

Metric	PoW	PoS	PoCmt
Gini	0.619 ± 0.107	0.693 ± 0.039	0.345 ± 0.005
Top-1% share	0.273 ± 0.163	0.171 ± 0.063	0.016 ± 0.000
Top-5% share	0.471 ± 0.144	0.413 ± 0.065	0.082 ± 0.001
HHI	0.080 ± 0.117	0.032 ± 0.017	0.004 ± 0.000

than PoW or PoS on every measured metric, with a Gini coefficient of 0.345 versus 0.619 and 0.693, and a top-1% share of 1.6% versus 27.3% and 17.1%.

IX. DISCUSSION

A. What PoCmt Changes

PoCmt does not change the structure of Byzantine consensus or the proof methodology. What it changes is the source of influence. In PoW and PoS, influence is a function of current holdings—hash power or stake—that can be acquired, pooled, and directed across identities at diminishing marginal cost. In PoCmt, influence is a function of accumulated participation history, which must be built per identity, per window, and cannot be pooled or transferred.

This shift has a concrete structural consequence: the cost of sustaining influence across s identities grows linearly in s rather than sublinearly. An entity that wishes to dominate consensus must supply proportionally more participation effort as it scales, rather than amortizing a fixed resource base. Whether this structural difference translates into greater decentralization in practice depends on deployment conditions—in particular on the properties of the participation resource and the economic cost of running participation processes—but the structural pressure toward concentration is removed.

TABLE VI
SUMMARY OF SIMULATION EXPERIMENTS.

Exp.	Variable	Key finding
1	Channel count m	Sharp transition at $m = s$; validates Theorem VII.1.
2	Identity count s	Share flat with s ; validates Corollary IV.4.
3	Bound τ	Majority reached at $m\tau = s$; motivates $\tau = 1$ design target.
4	Surplus ε	Share falls with ε ; validates Lemma VII.2.
5	Churn rate	Stable at $\leq 5\%$; validates Corollary VII.3.
6	Offline windows	Rapid recovery; validates Proposition V.4.
7	Gap $\Delta(d)$	Linear growth matches theory; validates Lemma VII.2.
8	PoW / PoS / PoCmt	PoCmt Gini 0.35 vs. 0.62 (PoW), 0.69 (PoS); top-1% share 1.6% vs. 27%, 17%.

B. What PoCmt Does Not Solve

PoCmt does not eliminate economic concentration. An adversary with sufficient resources can outsource participation to hired workers, automated farms, or cloud services. The adversarial model already accounts for this: the adversary operates m independent channels regardless of their implementation, and outsourcing changes the economic form of the cost without changing its asymptotic structure. Sustaining s identities still requires $\Omega(s)$ independent participation processes, each incurring a recurring per-window cost.

The difference from PoW and PoS is that this cost scales linearly with identity count and cannot be reduced through pooling. PoCmt shifts the bottleneck from capital accumulation to participation capacity, but does not make Sybil attacks free. As with all resource-based mechanisms [20], sufficiently wealthy adversaries remain capable of acquiring disproportionate influence; the mechanism removes the structural advantage that existing protocols provide to those who aggregate resources at scale.

C. Participation Markets and Outsourcing

A natural concern is that participation markets will emerge, enabling adversaries to purchase influence as easily as stake. This concern applies to any participation-based system and is not unique to PoCmt.

The critical distinction is structural: in a participation market, the buyer must still procure $\Omega(s)$ independent channel-windows to sustain s identities. There is no analogue to delegated staking, where a single entity can atomically direct an arbitrary pool of capital across many identities. Participation cannot be delivered in bulk from a single process; the per-process throughput ceiling τ applies regardless of whether processes are operated by the adversary directly or by contracted workers. The economic form of concentration may differ from PoW and PoS, but the structural scaling property is preserved.

D. Deployment Assumptions

The security analysis is conditional on Assumption IV.2: the deployed participation resource must satisfy (A1)–(A3) throughout execution. Verifying that a concrete challenge mechanism satisfies these properties under standard cryptographic assumptions is a deployment-time engineering problem not addressed here.

The bound τ in (A3) directly determines the quantitative security margin: a tighter τ limits adversarial capacity per process and strengthens honest majority. Experiment 3 (Section VIII-C) shows that adversarial leader share grows substantially as τ increases, motivating tight design of the participation resource. Different deployments may choose different trade-offs between τ , participation convenience, and infrastructure requirements.

E. Long-Range Attacks and Availability

The availability component U_v provides structural resistance to long-range attacks: dormant identities lose influence exponentially over time (Proposition V.4), preventing an adversary from accumulating participation history in one period and exploiting it much later. This property holds without checkpoint mechanisms, though checkpointing is compatible and would provide additional protection against reorganizations from before GST.

F. Open Problems

Concrete resource instantiation. The most immediate open problem is constructing a participation resource that provably satisfies (A1)–(A3) under standard cryptographic assumptions. Assumptions (A1) and (A2)—identity binding and window locality—are achievable via standard hash-based commitment and signature schemes. Assumption (A3)—bounded concurrent servicing—is the structural core, requiring that a single process cannot service more than τ distinct identities per window. Two enforcement approaches are available: *channel serialization* (per-session tokens, device-bound attestation, strict deadlines) and *cognitive serialization* for human-operated channels. Either approach suffices; deployments may select the mechanism best suited to their trust model. Proving (A3) with explicit τ under a formal adversarial solver model is left to future work.

Adaptive corruption. The security analysis assumes static corruption. Extending to adaptive corruption, where the adversary selects which validators to corrupt during execution, requires the Universal Composability framework and significantly increases proof complexity, following the approach of [18].

Incentive compatibility. Rational validators may deviate from honest participation if the expected benefit of free-riding exceeds the cost. A game-theoretic analysis following [20], [21] is needed to characterize equilibrium behavior and identify conditions under which honest participation is individually rational.

Privacy. Per-identity, per-window proofs generate participation transcripts that may reveal behavioral patterns or correlate

identities. Privacy-preserving resource designs using zero-knowledge proofs or anonymous credentials could address this while preserving (A1)–(A3).

X. CONCLUSION

Permissionless consensus protocols have traditionally derived influence from accumulated resources—computational power in Proof of Work, financial stake in Proof of Stake—creating structural pressure toward concentration that protocol rules alone cannot resolve. This paper has explored an alternative foundation: influence that emerges from sustained participation over time rather than from resource accumulation.

The core contribution is a participation-based model of consensus influence in which long-term commitment determines consensus power. We identified the structural properties that make participation non-aggregable, showed that sustaining influence across s identities over T windows requires $\Omega(sT)$ independent participation efforts regardless of capital or hardware, and constructed a complete permissionless consensus protocol on this foundation. Backbone-style safety (Theorem VII.4), liveness (Theorem VII.6), and commitment-proportional fairness (Lemma VII.7) hold under partial synchrony. Security derives from structural participation constraints, not from assumptions about computational scarcity, financial wealth, or human superiority over automation.

Seven analytical experiments confirm that adversarial influence scales with participation capacity rather than identity count, that commitment drift grows at the theoretically predicted rate, and that the protocol remains stable under moderate churn and transient offline periods.

Open problems include a concrete participation resource instantiation under standard cryptographic assumptions, incentive analysis under strategic validator behavior, extension to adaptive corruption via the UC framework, and privacy-preserving participation mechanisms. These questions notwithstanding, the results establish sustained participation as a structurally distinct and formally grounded basis for permissionless consensus—one that decouples influence from resource accumulation and offers a concrete alternative path toward long-term decentralization.

APPENDIX A

PROOF OF LEMMA VII.2 (QUANTITATIVE COMMITMENT DRIFT)

We give the full proof with explicit constants.

a) Setup.: Let $k_0 = \min_d k(d) \geq 1$, $\mu = \alpha\kappa_h k_0$ (per-channel per-window commitment increment), and $c = \tau/k_0$ (adversarial throughput factor). Let $n_{\mathcal{H}} = n_{\mathcal{H}}(d)$ and $m_{\mathcal{A}} = m_{\mathcal{A}}(d)$.

b) Honest commitment growth.: By Assumption (A3), every honest channel completes all $k(d) \geq k_0$ challenges per window after GST. Via (1): $H_v(d+1) - H_v(d) = \kappa_h k(d) \geq \kappa_h k_0$. The contribution to $W_{\mathcal{H}}$ via the α coefficient of (4) satisfies:

$$\mathbb{E}[W_{\mathcal{H}}(d+1) - W_{\mathcal{H}}(d) \mid \mathcal{F}_d] \geq \alpha\kappa_h k_0 \cdot n_{\mathcal{H}} = \mu \cdot n_{\mathcal{H}}.$$

c) Adversarial commitment growth.: By Proposition IV.3 (A3), the adversary's $m_{\mathcal{A}}$ processes generate at most $m_{\mathcal{A}} \cdot \tau$ valid proofs per window. Each proof contributes at most $\alpha\kappa_h$ to CS_v :

$$\mathbb{E}[W_{\mathcal{A}}(d+1) - W_{\mathcal{A}}(d) \mid \mathcal{F}_d] \leq \alpha\kappa_h \tau \cdot m_{\mathcal{A}} = \mu c \cdot m_{\mathcal{A}},$$

where $c = \tau/k_0 = O(1)$.

d) Net drift.:

$$\mathbb{E}[\Delta(d+1) \mid \mathcal{F}_d] \geq \Delta(d) + \mu(n_{\mathcal{H}} - c m_{\mathcal{A}}).$$

By Assumption III.4, $n_{\mathcal{H}} \geq (1 + \varepsilon)m_{\mathcal{A}}$, so:

$$n_{\mathcal{H}} - c m_{\mathcal{A}} \geq (1 + \varepsilon - c) m_{\mathcal{A}}.$$

Setting $\gamma = \mu(1 + \varepsilon - c)m_{\mathcal{A}} > 0$ completes the proof. Absorbing c into ε yields $\gamma = \mu\varepsilon m_{\mathcal{A}}$. $\square$

APPENDIX B

PROOF OF COROLLARY VII.3 (AZUMA–HOEFFDING CONCENTRATION)

a) Martingale construction.: Define $X_i = \Delta(D_0 + i) - \Delta(D_0) - i\gamma$ for $i \geq 0$. By Lemma VII.2, $\mathbb{E}[X_{i+1} \mid \mathcal{F}_{D_0+i}] \geq X_i$, so $\{X_i\}$ is a submartingale with $X_0 = 0$. Let $Y_i = -X_i$; then $\{Y_i\}$ is a supermartingale.

b) Bounded differences.: Each window changes Δ by at most B where $B = W_{\text{tot}}^{\max}$ is finite under bounded validator sets. Hence $|Y_{i+1} - Y_i| \leq B$ a.s.

c) Azuma–Hoeffding inequality.:

$$\Pr[Y_n \geq \xi] \leq \exp\left(-\frac{2\xi^2}{nB^2}\right).$$

Substituting $X_n = \Delta(D_0 + n) - \Delta(D_0) - n\gamma$ and setting $\xi = (d - D_0)\gamma/2$ yields:

$$\Pr\left[\Delta(d) \leq \Delta(D_0) + \frac{(d - D_0)\gamma}{2}\right] \leq \exp\left(-\frac{(d - D_0)\gamma^2}{2B^2}\right).$$

For $\Delta(D_0) > 0$, $\Pr[\Delta(d) \leq 0] \rightarrow 0$ exponentially. $\square$

APPENDIX C

PROOF OF THEOREM VII.4 (COMMON-PREFIX PROPERTY)

a) Post-stabilization leader probability.: By Corollary VII.3, $\exists D_1, \eta > 0$: for all $d \geq D_1$, $W_{\mathcal{H}}(d) \geq (1 + \eta)W_{\mathcal{A}}(d)$ with probability $1 - \exp(-\Omega(d))$. Hence for all epochs t in windows $d \geq D_1$:

$$p_{\mathcal{H}}(t) = \frac{W_{\mathcal{H}}(t)}{W_{\text{tot}}(t)} \geq \frac{1 + \eta}{2 + \eta} =: p_{\min} > \frac{1}{2}.$$

Set $p = p_{\min}$, $q = 1 - p_{\min}$, $\delta = p_{\min} - \frac{1}{2} > 0$. VRF pseudorandomness ensures leader elections are independent across epochs.

b) Fork probability via negative drift.: A conflicting fork of depth k requires the adversary to produce k blocks before the honest chain extends by k . By the ballot-problem argument:

$$\Pr[F_k] \leq \left(\frac{q}{p}\right)^k \leq e^{-2\delta k} = \text{negl}(k).$$

c) *Pre-stabilization epochs.*: For $d < D_1$, the number of such epochs is finite. Common-prefix for these blocks follows from the standard backbone analysis of [3], [17] under (A3).

d) *Slashing reinforcement.*: Detected equivocation triggers $P_v \leftarrow \delta_s P_v$ ($\delta_s < 1$), reducing W_A and tightening the exponential bound.

e) *Conclusion.*: A common-prefix violation at depth k requires F_k , which occurs with probability $\leq e^{-2\delta k} = \text{negl}(k)$. $\square$

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